Conservation of Energy

- Work - kinetic energy theorem

$$K_f - K_i = W \qquad \left(\text{proof by } W = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}\cdot d\vec{r},\quad \vec{F} = m\vec{a}\right)$$

rewriting:

$$K_f - K_i \underbrace{- W}_{= U_f - U_i} = 0$$

$U_f - U_i$: change of "potential energy"

$$\Rightarrow \quad K_f + U_f = K_i + U_i$$

i.e. Conservation of mechanical energy (机械能守恒):

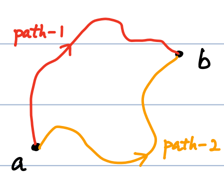
$$E_{mec} \equiv K + U \quad \text{is constant}$$

- requirement

Conservative force (保守力):

Work done around any closed path (round trip) is zero. $\boxed{\text{definition}}$

With a conservative force,



$$W_{ab,1} = W_{ab,2}$$

Work depends ONLY on Initial & Final Positions.

Proof:

$$\begin{cases} W_{ab,1} + W_{ba,2} = 0 & (\text{according to def}) \\ W_{ba,2} = -W_{ab,2} & (W = \int \vec{F}\cdot d\vec{r}) \end{cases}$$

$d\vec{r}$ opposite for $a \to b$ & $b \to a$)

$$\Rightarrow \quad W_{ab,1} = W_{ab,2}$$

A further look at

$$K_f - K_i \underbrace{- W}_{= \, U_f - U_i} = 0$$

$-W \equiv U_f - U_i$ depends only on $U_f - U_i$

$\Leftrightarrow$ consistent with definition of conservative force.

- How to compute $\Delta U = U_f - U_i$?

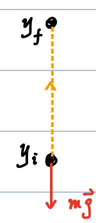
$$\Delta U = -W$$
$$= - \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}(\vec{r}) \cdot d\vec{r}$$

- <u>Gravitational potential energy</u>

$$U_f - U_i = \Delta U = - \int_{y_i}^{y_f} (-mg)\, dy$$
$$= mg\,(y_f - y_i)$$

$$\Rightarrow \begin{cases} U_f = mg\,y_f + C \\ U_i = mg\,y_i + C \end{cases}$$

$$\Rightarrow \quad U(y) = mg\,y + \overset{\text{constant}}{C}$$

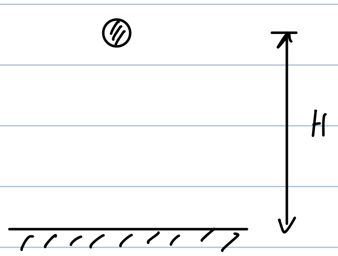


Notes:

- C is arbitrary constant (fixed during calculation)

  often set $C = 0$ (not necessary)
- No dependence on horizontal displacement.
- assuming earth is static ($m_{earth} \gg m_{object}$)
- assuming close enough to surface of earth.

Example.



$t_i = 0$, $y_i = H$, $v_i = 0$,

$K = \frac{1}{2} m v_i^2 = 0$

$U = mgH$

$\Rightarrow \quad E_{mec}^{(i)} = K + U = mgH$

$$t_f = \sqrt{\frac{2H}{g}}, \quad y_f = 0, \quad v_f = \sqrt{2gH}$$

$$\begin{cases} K = \frac{1}{2} m v_f^2 = mgH \\ U = mg y_f = 0 \end{cases}$$

$$\Rightarrow \quad E_{mec}^{(f)} = K + U = mgH$$

$$\Rightarrow \quad \boxed{E_{mec}^{(i)} = E_{mec}^{(f)}}$$

(Energy transforming from U to K)

- Example (opposite process)

obviously, again,

$$E_{mec}^{(i)} = E_{mec}^{(f)}$$

(Energy transforming from K to U)

- Elastic potential energy

$$U_f - U_i = \Delta U = -\int_{x_i}^{x_f} (-kx)\, dx$$

$$= \frac{1}{2} k \left(x_f^2 - x_i^2 \right)$$

$$\Rightarrow \quad U(x) = \frac{1}{2} k x^2 + \underset{\text{often set } C=0}{C}$$

With $C = 0$ convention,

$$U(x) = \frac{1}{2} k x^2$$

Notes:

- $\begin{cases} U(0) = 0 & \text{undeformed} \\ U(x \neq 0) > 0 & \text{deformed (positive energy stored)} \end{cases}$

- Calculate conservative force.

(previously, given $F \Rightarrow \Delta U = -\int_{x_i}^{x_f} F(x)\, dx$)

Now, given $U(x)$,

$$\Delta U(x) = -W = -F(x)\,\Delta x \qquad (\Delta x \to 0)$$

$$\Rightarrow \quad \boxed{F(x) = -\frac{dU(x)}{dx}}$$

Check 1. gravitational potential

$$U(x) = mgx$$

$$\Rightarrow \quad F(x) = -\frac{dU(x)}{dx} = -mg \qquad \checkmark$$

check 2. elastic potential.

$$U(x) = \frac{1}{2}kx^2$$

$$\Rightarrow \quad F(x) = -\frac{dU(x)}{dx} = -kx \qquad \checkmark$$

- higher dimension

$$\Delta U = -\int_{\vec{r}_i}^{\vec{r}_f} \vec{F}(\vec{r}) \cdot d\vec{r}$$

$$\vec{F}(\vec{r}) = -\nabla U(\vec{r})$$

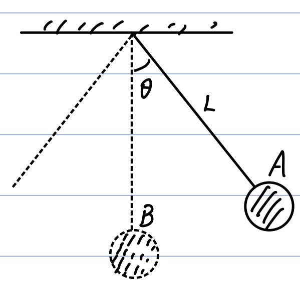
In cartesian coordinates

$$\nabla \equiv \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

i.e. $\vec{F} = -\nabla U$

$$= -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$$

Example.

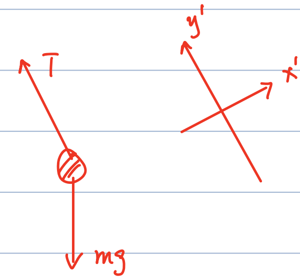


- Released at A with $V_A = 0$

$$0 + mg(L - L\cos\theta) = \frac{1}{2}mv_B^2 + 0$$

$$\Rightarrow \quad v_B = \sqrt{2gL(1-\cos\theta)}$$

- Q: $a_A = ?$, $T_A = ?$



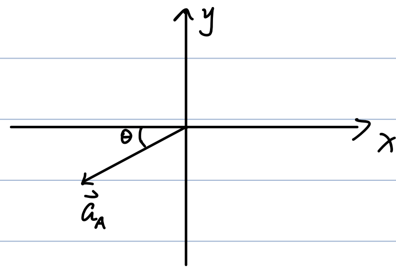
y'-dir: $T_A - mg\cos\theta = m\frac{0^2}{r} = 0$

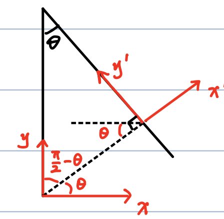
$$\Rightarrow \quad T_A = mg\cos\theta$$

x-dir: $\quad -mg\sin\theta = m a_A$

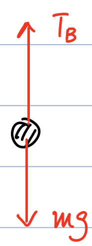
$\Rightarrow \quad a_A = -g\sin\theta$

$$\begin{cases} a_A^x = -g\sin\theta \cdot \cos\theta \\ a_A^y = -g\sin\theta \cdot \sin\theta \end{cases}$$





- Q: $a_B = ?$ $\quad T_B = ?$

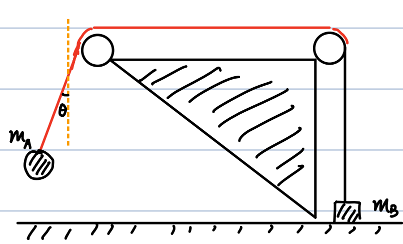


$$a_B = \frac{v_B^2}{L} = \frac{2g\cancel{L}(1-\cos\theta)}{\cancel{L}} = 2g(1-\cos\theta)$$

$$T_B - mg = m a_B$$

$$\Rightarrow \quad T_B = mg(3 - 2\cos\theta)$$

Note: $\quad T_A < T_B \quad$ for $\quad 0 < \theta \leq \frac{\pi}{2}$



Q: given m_A, m_B, what's maximal value of θ, such that m_B stays on ground?

sol: we already know

$$T_B = m_A g(3 - 2\cos\theta) \qquad \text{maximal tension}$$

$$m_A g(3 - 2\cos\theta) \leq m_B g$$

$$\Rightarrow \quad 3 - 2\cos\theta \leq \frac{m_B}{m_A}$$

$$\Rightarrow \quad \cos\theta \geq \frac{3m_A - m_B}{2m_A}$$

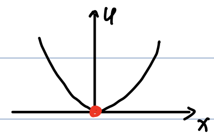
- Equilibrium

  - stable equilibrium :

    minimum of $U(x)$

    $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} > 0$

    e.g. $U(x) = \frac{1}{2}kx^2$



$x = 0$ (undeformed)

Near $x=0$

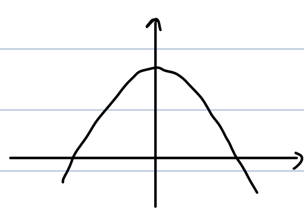
$F(x) = -kx$ (points to $x=0$)

  - Unstable equilibrium :

    maximum of $U(x)$

    $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} < 0$

    e.g. $U = -\frac{1}{2}kx^2$



near $x=0$

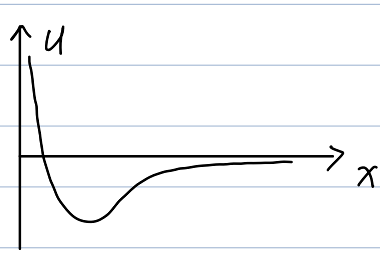
$F = -\frac{dU}{dx} = kx$

(points away from $x=0$)

Example. Lennard-Jones potential

(potential energy of 2 neutral atoms in a moleculer)

$U(x) = 4\varepsilon\left[\left(\frac{\sigma}{x}\right)^{12} - \left(\frac{\sigma}{x}\right)^{6}\right]$



Stable equilibrium position:

$$\frac{dU}{dx} = 0 \Rightarrow \quad -12\frac{\sigma^{12}}{x^{13}} + 6\frac{\sigma^6}{x^7} = 0$$

$$\Rightarrow \quad x = \sqrt[6]{2}\,\sigma$$

$$\frac{d^2U}{dx^2} = 4\varepsilon\left[12\times13\frac{\sigma^{12}}{x^{14}} - 6\times7\frac{\sigma^6}{x^8}\right]$$

$$\overset{x=\sqrt[6]{2}\sigma}{=\!=} 4\varepsilon\left[156\frac{\sigma^{12}}{2^{\frac{14}{6}}\sigma^{14}} - 42\frac{\sigma^6}{2^{\frac{8}{6}}\sigma^8}\right]$$

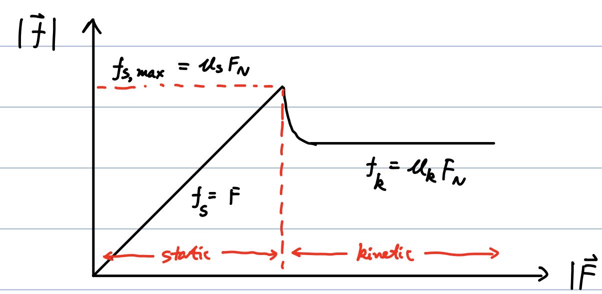
$$= 36\sqrt[3]{4}\frac{\varepsilon}{\sigma^2} > 0 \quad \Rightarrow \text{stable}$$

- Non conservative force.

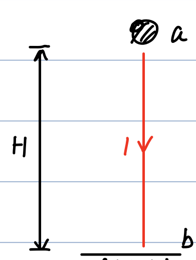
force that does NOT satisfy definition of "conservative force"

e.g. kinetic friction, drag, ...

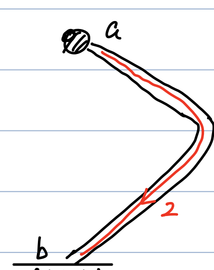
remind: friction



Example: work done by friction.



$W_{ab,1} = 0$



$W_{ab,2} = \int \vec{f}_k \cdot d\vec{l} \neq 0$

$W_{ab,1} \neq W_{ab,2} \quad \Rightarrow$ Non-conservative force

$\Rightarrow$ No conservation of mechanical energy

Path-1 : $E^a_{mec} = E^b_{mec} = mgH$

Path 2 : $E^a_{mec} = mgH$

$E^b_{mec} = 0$ (assume at b, $v=0$)

Total energy is conserved!

$E_{tot} = E_{mec} + E_{thermal}$

(thermal energy or internal energy)

↓

internal kinetic energy of atoms and molecules

More on energy :

- $E = mc^2$ mass is also energy (relativity)
- E discretized on microscopic scale (quantum mechanics)